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Expansion of açaí fruit and heart-of-palm production promotes hybridization and introgression in *Euterpe* palms

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Abstract

Background Palms of the genus *Euterpe* have multiple uses and hold great ecological and socioeconomic importance. *Euterpe edulis* is a key species in the Atlantic Forest, providing food for many animal species. The açaí palms, *E. oleracea* and *E. precatoria*, are essential in traditional Amazonian food and medicine. However, increasing demand for açaí fruit and heart-of-palm has led to the expansion of *E. oleracea* plantations into both the Amazon and the Atlantic Forest, raising concerns about hybridization and introgression among species.

Results Using single-nucleotide polymorphism (SNP) markers, we found that putative *Euterpe* hybrids exhibit higher genomic diversity than their parental species. Population structure analyses revealed that these individuals occupy intermediate positions between species and form nearly exclusive genomic clusters. Evidence of introgression was also detected, particularly at *E. oleracea* introduction sites. Niche overlap analysis indicated that *E. oleracea* has high potential to expand into the niche of *E. edulis*, while maintaining niche stability with *E. precatoria*. Niche models further indicated broad environmental suitability for *E. oleracea* from the Atlantic Forest coast to the entire Amazon region.

Conclusions We confirm the occurrence of hybridization and introgression among *Euterpe* species. The identification of hybridization zones between *E. oleracea* and *E. edulis* in the Atlantic Forest highlights the urgency of conservation strategies for *E. edulis*. In the Amazon, it is crucial to develop new cultivars with non-overlapping flowering and fruiting periods. Additionally, monitoring spontaneous occurrences is essential to prevent genetic contamination of natural populations.

Keywords Tropical, Arecaceae, Ecological niche modeling, Niche overlap, Genotyping by sequence, SNPs

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Background

Hybridization can occur naturally when interspecific incompatibility barriers are overcome during secondary contact between congeneric species, resulting in viable offspring [1]. This process can be facilitated by shared seed dispersers, synchronized flowering periods, and reduced geographic distance [2, 3], and may increase the occurrence of spontaneous hybrids at the site of introduction [4]. Introgression, in turn, refers to the incorporation of alleles from one species into the gene pool of another [5]. While hybridization represents the initial formation of interspecific individuals, introgression describes the longer-term genetic exchange that follows, with significant implications for evolution and conservation. On the one hand, introgression may introduce novel genetic variation that enhances adaptive potential, potentially leading to the formation of genotypes more resistant to biotic or abiotic stressors in newly colonized environments [6]. On the other hand, it may cause maladaptive genetic changes, the extinction of pure parental lineages, or the emergence of highly invasive hybrid forms [7]. These processes can pose a serious threat to the integrity of native genetic resources, potentially eroding locally adapted gene pools and undermining conservation efforts aimed at preserving wild populations.

Palms are key species in the ecosystems where they occur and hold significant socioeconomic importance [8]. All parts of a palm can be utilized, resulting in a wide range of economic applications including food, medicine, construction materials, and ornamental uses [9]. In Brazil, three species of the genus *Euterpe* (subfamily: Arecoideae; tribe: Euterpeae) are of particular ecological and socioeconomic importance: *Euterpe edulis* Mart., *E. oleracea* Mart., and *E. precatoria* Mart.

Popularly known as "juçara", *E. edulis* is a single-stemmed, ecologically keystone palm species in the Atlantic Forest biome of Brazil, as its fruits serve as an important food source for both mammals and birds [10]. The species produces one of the most valued nontimber forest products: the heart-of-palm [11]. The açai palms are represented by *E. oleracea* and *E. precatoria*. *E. oleracea* is a multistemmed species whose fruits have long been used in local diets and traditional medicine [12]. In Brazil, it occurs naturally in the eastern portion of the Amazon biome [13], where it is gradually replaced by *E. precatoria* in the western Amazon basin. *E. precatoria*, a single-stemmed palm, is essential for food, medicine, and building materials in the region [12]. In the Pacific coastal regions of Colombia and Ecuador, the two species occur sympatrically [13].

Between 1986 and 2011, the average annual production of açai fruit in Brazil was 117,063.59 tonnes. From 2011 to 2018, production rose to 211,100.63 tonnes per year, representing an 80.3% increase in yield [14]. The state of Pará is the largest producer of açaí fruit (58.4%), followed by Amazonas (31.0%). Until 2004, açaí production in Amazonas was based almost entirely (98%) on the extraction of naturally occurring *E. precatoria* palms [15]. Since then, BRS-Pará, a productive *E. oleracea* cultivar, has been introduced in drier areas. This cultivar enables year-round production, as the peak harvest period of *E. precatoria* occurs between March and July, whereas *E. oleracea* is harvested mainly from September to December [6].

Using four nuclear low-copy markers and a plastid region, Pichardo-Marcano et al. [16] determined that most species in tribe Euterpeae diverged within the last 10 million years and colonized the Caribbean approximately 2 Mya. *E. edulis* is thought to have diverged from its common ancestor through vicariant speciation approximately 1.4 Mya [16]. As *E. edulis* is a single-stemmed species, harvesting requires cutting the entire plant, and the overexploitation has led to species to be listed as "vulnerable" in the Brazilian Flora Red Book [17]. To reduce pressure on natural populations, controlled hybridization with the multistemmed *E. oleracea* was initiated in the 1970s [3]. This also promoted the introduction of *E. oleracea* into parts of southeastern Brazil, where [3] natural hybrids between with *E. edulis* have since been documented [4]. More recently, the expansion of *E. oleracea* plantations into the Atlantic Forest biome, driven by the growing demand for açaí pulp, aims to bring production closer to consumption centers and reduce logistical challenges. However, this geographic and biome shift raises biological concerns regarding hybridization and introgression with the already vulnerable *E. edulis* [4].

Anthropogenic activities have been the main drivers of *E. oleracea* dispersal beyond its natural range in the Amazon biome, particularly through cultivation in central Amazonian states such as Amazonas. In contrast, *E. precatoria* remains predominantly exploited through extractivism within its natural range [15]. These congeners share a mixed mating system, with both self and cross-pollination occur, although allogamy predominates [18–20], making interspecific crossing possible. Reports of hybrids between them are relatively recent and may be linked to the expansion of *E. oleracea* plantations into regions where *E. precatoria* is native. Putative hybrids have been identified based on the observation of intermediate phenotypes and nuclear DNA content values between the two parental species, as measured by flow cytometry [21, 22]. More recently, population genomics has enabled the identification of thousands of markers to investigate gene flow and hybridization between species [23]. Genomic parameters can help distinguish hybrids from parental species, as hybrids often exhibit increased genetic diversity, including increased heterozygosity [24]. In this context, single nucleotide polymorphisms (SNPs), highly abundant markers with broad genomic coverage [25, 26], provide a powerful tool to assess population structure and detect hybrids by evaluating genetic distances and allele frequency patterns [27]. Moreover, despite hybridization and introgression among *Euterpe* palms appear to be strongly influenced by human activities, it is essential to characterize their ecological niches and identify potential natural hybrid zones to evaluate the extent to which these processes may occur independently of anthropogenic influence.

Ecological niche modeling (ENM) is based on quantifying species-environment relationships and on the assumption that ecological niches are conserved of across space and time [28]. This approach assumes that a species can establish itself in environments with conditions similar to those of its native range [29], which may be the case for the spread of *E. oleracea*. It is therefore important to distinguish the niche-related mechanisms that allow a species to establish in new regions [29]. To understand how the niche of *E. oleracea* overlaps with those of *E. edulis* and *E. precatoria*, it is necessary to quantify niche stability (i.e., the overlap between the niche of *E. oleracea* and those of the other *Euterpe* species), niche unfilling (the portions of *E. edulis* and *E. precatoria* niches not yet colonized by *E. oleracea*), and niche expansion (the portion of *E. oleracea*'s niche that is not shared with the other species), as described by Pearman et al. [28, 30]. In addition, ENM can be used to geographically identify areas with high suitability for each species, as well as regions of overlap. This makes it possible to pinpoint areas of potential secondary contact, which may represent zones of natural hybridization [7].

Thus, in this study, we propose the integration of population genomics, niche overlap analysis, and ecological niche modeling to address four main objectives. First, we aimed to demonstrate hybrid formation through population genomic parameters. Given that controlled hybridization experiments have been conducted (*E. oleracea* × *E. edulis*) and that putative hybrids have already been visually identified (*E. oleracea* × *E. edulis*; *E. precatoria* × *E. oleracea*), we expect that these individuals will exhibit greater genomic diversity than the parental species. Second, we aimed to define patterns of population structure that reflect hybridization and introgression between congeneric species. In hybrids, genomic clusters may differ from those of the parental species, showing exclusive or intermediate ancestry. In addition to the occurrence of natural hybrids, introgression may result in the incorporation of alleles between parental species in sympatry. Third, we aimed to quantify the niche overlap between *E. oleracea* and the other *Euterpe* species to understand their ecological interactions. The niche of *E. oleracea* is expected to expand, especially compared with that of *E. edulis*, because of its recent introduction into the Atlantic Forest. In contrast, the niche overlap between *E. oleracea* and *E. precatoria*, both native to the Amazon biome, is expected to reflect greater stability, with little change in their environmental distributions. Fourth, we aimed to predict geographic areas where hybridization and introgression are most likely to occur. We expect that açai palms will remain more suitable for the Amazon biome, whereas *E. oleracea* and *E. edulis* will be distributed along the coastal Atlantic Forest, where their occurrence has already been documented.

Materials and methods

Population genomics

Sampling

To investigate hybridization and introgression among species of the genus *Euterpe*, leaf samples were collected from all three species (*E. edulis*, *E. oleracea*, and *E. precatoria*), as well as from hybrids of *E. oleracea* × *E. edulis* and *E. precatoria* × *E. oleracea*. These samples were organized into two distinct datasets for analysis. To date, no evidence of hybridization between *E. edulis* and *E. precatoria* has been reported.

The first dataset included individuals of *E. edulis*, *E. oleracea*, and hybrids of *E. oleracea* × *E. edulis*. For *E. edulis*, leaf samples from 54 individuals were collected at five sites in the state of São Paulo between 2015 (SISGEN number AFD6956) and 2021 (SISGEN number A411583). Three of these sites were located in Ubatuba (Casa da Farinha, Fazenda Capricórnio, and IAC Ubatuba), where the introduction of *E. oleracea* into the Atlantic Forest biome has been reported [3, 4]. The other two sites, Ilha do Cardoso and Santa Virgínia, have no documented records of *E. oleracea* introduction. The *E. edulis* samples were collected by Maria I. Zucchi and Ana F. Francisconi, with formal identification confirmed by Maria I. Zucchi. For *E. oleracea*, 43 accessions were included from four municipalities in Pará state (Acará, Inhangapi, Peixe-Boi, and São Francisco do Pará). The *E. oleracea* samples were collected and identified in 2016 (SISGEN number A411583) by Santiago L. F. Ramos. The *E. oleracea* × *E. edulis* hybrids were derived from a controlled cross performed by Bovi et al. [3] using *E. edulis* (male parent) from Instituto Agronômico de Campinas (IAC), Ubatuba and *E. oleracea* (female parent) introduced from Pará in approximately 1950. Previous evaluations demonstrated that these hybrids presented enhanced vegetative growth and higher heart-of-palm yield relative to the parental species, while also displaying considerable variability in agronomic traits and reduced fertility. Subsequent backcrosses resulted in only a few multi-stemmed individuals, indicating that this trait may follow a maternal inheritance pattern [3, 4]. These individuals are currently planted at Fazenda Santa Cecília, IAC, São Paulo (Fig.1). Leaf material from 16 of these hybrids was collected in 2021 by Valeria A. Modolo and

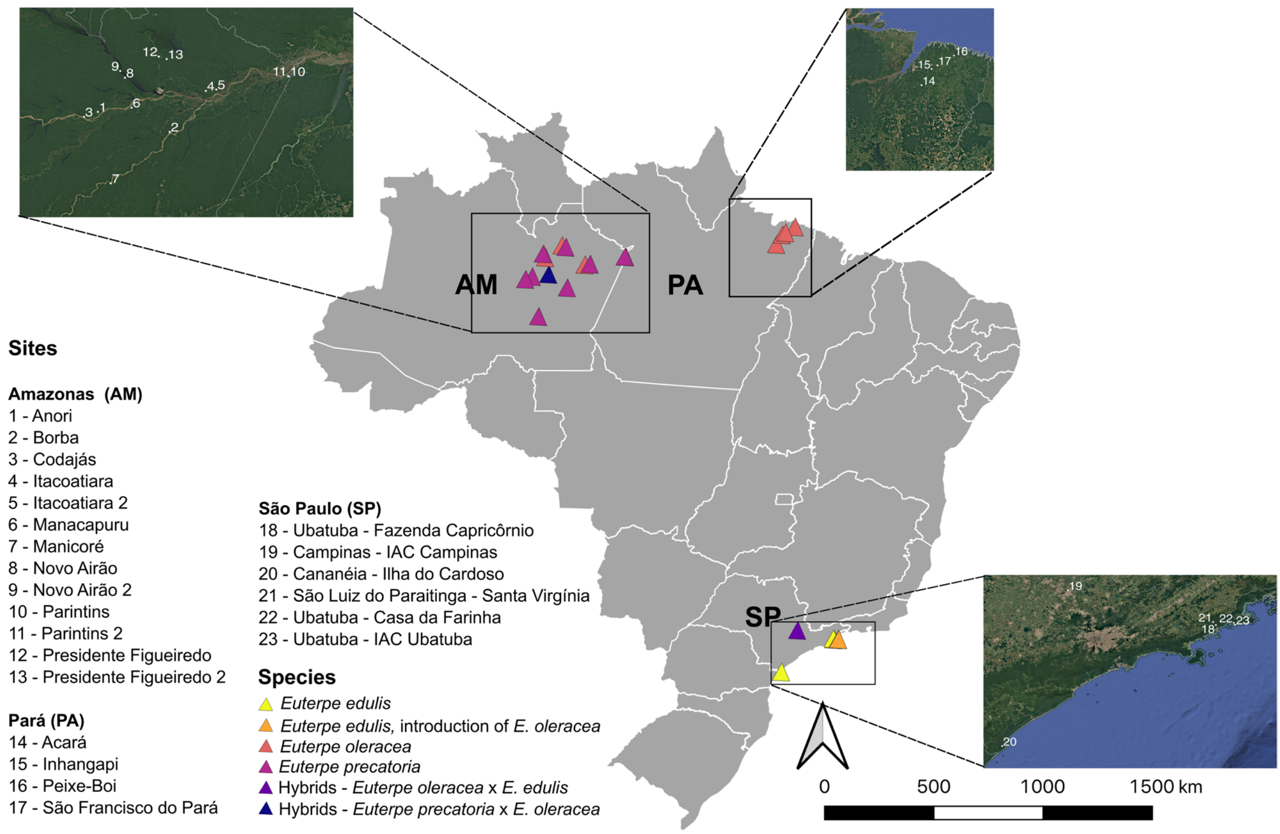

Fig. 1 Geographic locations of the sampled individuals used in this study. In dataset 1, *E. oleracea* was sampled from four sites in Pará state (PA; 43 individuals), *E. edulis* from five sites in São Paulo state (SP; 54 individuals), and *E. oleracea* × *E. edulis* hybrids from a plantation in Campinas, SP (16 individuals). In dataset 2, *E. oleracea* was sampled from four sites in Amazonas state (AM; 50 individuals) and four sites in Pará state (PA; 43 individuals). *E. precatoria* was sampled from eight sites in Amazonas (AM; 114 individuals). Putative *E. precatoria* × *E. oleracea* hybrids were sampled from a plantation in Manacapuru, AM (5 individuals)

Jonathan A. Morales-Marroquín, identified by Valeria A. Modolo (SISGEN number A411583).

The second dataset included individuals of *E. oleracea*, *E. precatoria*, and putative hybrids of *E. precatoria* × *E. oleracea*. Leaf material was obtained from 93 individuals of *E. oleracea* and 114 of *E. precatoria*. The *E. oleracea* samples included 43 accessions from Pará, collected and identified in 2016 by Santiago L. F. Ramos (as in dataset 1), and 50 individuals from Amazonas, collected and identified in 2016 by Maria T. G. Lopes. The *E. precatoria* samples were collected and identified across eight sites in Amazonas in 2015 by Santiago L. F. Ramos. In four municipalities—Itacoatiara, Novo Airão, Parintins, and Presidente Figueiredo—both açaí species were collected. The putative hybrids of *E. precatoria* × *E. oleracea* consisted of five individuals identified within an *E. precatoria* plantation in Manacapuru, Amazonas (Fig. 1). According to local reports, all individuals were reportedly propagated from a single *E. precatoria* matrix. However, some individuals presented tillering patterns and leaf morphologies distinct from those of typical *E. precatoria*, suggesting that they may be natural hybrids. In addition, flow cytometry analyses revealed nuclear DNA content values intermediate between the two parental species, further supporting the hypothesis that they represent natural hybrids [21]. Putative hybrid samples were obtained and identified in 2019 by Maria T. G. Lopes (SISGEN number A411583). Voucher specimens were not deposited in a public herbarium, as leaf tissue was collected specifically for molecular analyses and preserved in silica gel.

DNA extraction and genotyping-by-sequencing library

Genomic DNA was isolated from adult leaf tissue using a 2% CTAB protocol [31]. DNA concentration and quality were assessed both by electrophoresis on 1% agarose gels stained with GelRed™ (Biotium, Hayward) and with a Qubit fluorometer (Thermo Fisher Scientific, USA). SNP markers were generated through a genotyping-by-sequencing (GBS) library, which employs restriction enzymes to reduce genome complexity and adapters with barcodes to enable multiplexing for next-generation sequencing. In our study, genomic DNA was digested with the restriction enzymes MseI and PstI in double-digest protocol following Poland et al [32].

Sample-specific barcodes were ligated to the fragments for individual identification, and the resulting libraries were PCR-amplified using primers complementary to the adapters. Sequencing was carried out on the Illumina NextSeq 2000 platform, producing single-end reads. On average, 0.66 million reads were generated per sample.

Verification of the quality of sequencing, filtering, and obtaining SNP markers

For both datasets, FastQC [33] was used to assess the overall quality of the raw sequencing reads. The Stacks 2.62 pipeline [34] was then applied separately to each dataset to demultiplex the samples based on barcodes, perform filtering, generate a catalog of loci, and call SNPs.

The first module applied was *process_radtags*, which demultiplexed the samples based on their barcodes and applied quality filtering. The filters included minimum quality score thresholds (-q), removal of reads containing uncalled bases, and correction of overrepresented sequences. Multiqc [35] was used to summarize the FastQC results per individual, allowing quality assessment after demultiplexing. Since there is no reference genome available for the genus *Euterpe*, the *ustacks* module was run de novo. The parameter m = 4 was used as the minimum read depth required to form a stack. The number of nucleotide mismatches allowed between stacks within individuals was set to M = 3, and the number of secondary reads allowed was N = 5. The catalog was built via *cstacks* by identifying consensus sequences for each SNP, with the maximum number of mismatches between loci across individuals (n) set to the same value as M. Population-level filtering was performed using the *populations* module, with a minor allele frequency (min−maf) threshold of 1%, requiring the allele to occur in at least 75% of individuals within a population (r = 0.75) and to be present in all but one population. We did not restrict the dataset to putatively neutral loci, since our objective was to capture genome-wide patterns of divergence, hybridization, and introgression, including loci potentially under selection. We acknowledge that the inclusion of such loci may accentuate estimates of genomic differentiation but excluding them could also bias inferences by omitting loci involved in the processes under study. Moreover, selection in hybrids is context-dependent, varying with demographic history and ecological environment [36].

Analysis of genomic diversity

Each dataset was characterized based on estimates of genomic diversity. In dataset 1, statistics were calculated separately for each collection site of *E. edulis* and *E. oleracea*, as well as for the controlled hybrids (*E. edulis* × *E. oleracea*). In dataset 2, estimates were obtained for each collection site of *E. precatoria* and *E. oleracea*, and for the putative hybrids (*E. precatoria* × *E. oleracea*). The observed heterozygosity and gene diversity (H_O and H_S), allelic richness (A_R), fixation index (f), along with their 95% confidence intervals estimated from 1,000 bootstrap replicates, were calculated via the *hierfstat* 0.5.11 R package [37]. The numbers of alleles (A) and private alleles (A_P) were determined using the *Adegenet* 2.1.6 [38] and *poppr* 2.9.6 R packages [39].

Population structure

Population structure analyses were conducted separately for each dataset to investigate hybridization and introgression between congeneric species. Wright's F-statistics (F_{ST}, F_{IT}, F_{IS}) were calculated for each dataset, considering putative parental species and hybrids, using the *hierfstat* R package [37]. Pairwise F_{ST} values, which are based on Nei's genetic distance [40], were also computed according to the sampling site. The results were visualized via heatmaps generated with the *pheatmap* 1.0.12 R package [41].

To investigate hybridization and introgression based on patterns of genomic structure, we applied three structuring methods. First, two multivariate analyses were performed. (1) Principal component analysis (PCA) was conducted via the *ade4* 1.7–22 R package [42] to assess how individual genetic variability is distributed in multivariate space. (2) Discriminant analysis of principal components (DAPC) was performed using the *adegenet* R package. This method combines PCA with discriminant analysis (DA) to identify genomic groups, maximizing between-group variation while minimizing within-group variation [43]. An unsupervised *K-means* clustering approach was used to determine the number of genetic clusters prior to DAPC.

The remaining method was (3) sparse nonnegative matrix factorization (sNMF), which is likelihood-based algorithms designed for inferring population structure [44]. sNMF solves a constrained least-squares minimization problem via an alternating algorithm that is based on the block principal pivoting method [45]. This analysis was performed in the *LEA* 3.16.0 R package [46]. For dataset 1 (*E. oleracea*, *E. edulis*, and their hybrids), we tested values of K from 1 to 10. For dataset 2 (*E. oleracea*, *E. precatoria*, and their hybrids), which included a relatively large number of sampling sites, K values from 1 to 17 were tested. We used cross-entropy criterion to select the optimal model in sNMF and visualize the simulation results. In addition, admixture bar plots for K = 2–6 (dataset 1) and K = 2–8 (dataset 2) were provided as Supplementary Figures to investigate structure patterns across different K values.

Niche overlap and ecological niche modeling

Data collection

The occurrence data for the palms *E. edulis*, *E. oleracea* and *E. precatoria* were compiled from multiple databases (Supplementary Fig. 1). These included the Center for Tropical Forest Science (https://forestgeo.si.edu/), GBIF (https://www.gbif.org/), Missouri Botanical Garden (https://www.missouribotanicalgarden.org/), SALVIAS (https://bien.nceas.ucsb.edu/bien/salvias-update/), speciesLink (http://splink.cria.org.br/), National Herbarium Nederland (http://www.nationaalherbarium.nl/), New York Botanical Garden (https://www.nybg.org/), Wildlife Insights (https://www.wildlifeinsights.org/team-network), and Forest Global Earth Observatory (https://forestgeo.si.edu/) using the *BIEN* 1.2.6 ("BIEN_occurrence_species") and *spocc* 1.2.3 ("occ" and "occ2df") [47] R packages. Additional records were obtained from Brazilian databases, including Portal da Biodiversidade (https://portaldabiodiversidade.icmbio.gov.br/portal) and Sistema de Informação sobre a Biodiversidade Brasileira SiBBr (https://www.sibbr.gov.br), as well as from the literature sources on Atlantic flower-invertebrates [48] and Atlantic frugivory [49].

After data retrieval, initial coordinate filtering was performed using the *CoordinateCleaner* [50] (Supplementary Fig. 1). The following criteria were applied: (i) missing data filter, which excluded occurrences lacking longitude and/or latitude information, as well as duplicates, zero coordinates, or identical values; (ii) spatial filter, which retained only coordinates located outside capital cities (2,000 m buffer), country and municipal centroids (2,000 m), GBIF headquarters, biodiversity institutions (100 m), ocean areas, and urban zones; (iii) outlier detection, which uses the quantile method with a minimum of five records to flag spatial outliers; and (iv) temporal filtering, which excluded records dated before 1980 to minimize uncertainty in geographic coordinates, since older records generally have lower spatial accuracy due to the absence of GPS and less standardized locality descriptions, and to filter for possible nomenclature or geolocation errors. For *E. oleracea*, additional manual curation was performed to remove cultivated accessions based on associated metadata. After filtering, one occurrence point was added for each sampling site included in the population genomics study (available at: https://github.com/anaf-f/Euterpe_hybrids). Finally, a minimum of five occurrences per species was needed, and spatial rarefaction was applied based on the resolution of the environmental layers (~ 10 km). These parameters used the arguments "min_occ" and "thin_occ" from the *ENMTML* 1.0.0 R package [51].

Niche overlap

To explore the niche overlap of the *Euterpe* species, we performed an environmental space reduction. The reduction was based on an ordination technique with 19 bioclimatic and one solar radiation sources from WorldClim 2.1 [52] at a resolution of 2.5 arcminutes (~ 5 km) (PCA-env sensu) [53]. In addition to the bioclimatic variables, eight soil predictors were also added to the palm distribution models. These variables included the volume fraction of coarse fragments; organic carbon density; pH (H_2O); and clay, nitrogen, sand, silt, and soil organic carbon contents obtained from SoilGrids (www.soilgrids.org) [54]. These predictors were adjusted to the same resolution as the bioclimatic variables. A simple average was calculated for the topsoil layers up to 30 cm deep, which represent the main root zone of *E. edulis* [55], and this depth was also considered appropriate for *E. oleracea* and *E. precatoria*.

The PCA-env was calibrated via the climate and edaphic information from pairwise combinations of the three species (*E. oleracea - E. edulis* and *E. oleracea - E. precatoria*). The first two PCA-env axes were divided into 100 × 100 cells, spanning the full range of observed values, using the function "ecospat.grid.clim.dyn" from the *ecospat* 4.1.2 R package [56].

To determine the overlap between species, Schoener's D metric [57, 58] was quantified as proposed by Broennimann et al. [53], using the function "ecospat.niche.overlap" from the *ecospat* R package [56]. This metric ranges from 0 to 1, indicating total dissimilarity or complete niche overlap, respectively [59]. Additionally, the percentages of niche expansion, stability, and unfilling among the *Euterpe* species were calculated based on the same function [56, 59].

Ecological niche modeling

In the pre-processing phase of the models, we used the set same bioclimatic variables, solar radiation, and soil predictors with a resolution of 2.5 arcminutes (~ 5 km) described above. We performed a PCA to reduce the collinearity among the variables, and a correlation matrix derived from the PCs was generated. A new set of PCA-derived variables was selected based on the criterion of retaining PCs that explained at least 95% of the total variance [60]. This methodology was also implemented with the *ENMTML* R package ("colin_var = c(method='PCA')") [51]. The variables represented the entire Neotropical region [61]; however, the model fitting was restricted by a buffer defined by the maximum pairwise distance between occurrences for each species. This was specified using the "sp_accessible_area" argument with the "BUFFER" method of the *ENMTML* R package.

The "env_const" approach was employed to determine pseudoabsences in the *ENMTML* R package [51], which was restricted to environmentally dissimilar regions with lower suitability values predicted by a Bioclim model [62]. To improve model accuracy, the data were partitioned into 70% for training and 30% for testing. The bootstrap method was also applied, with the procedure repeated 10 times for each species.

In the processing phase, we used different algorithms to estimate the ecological niche of the species, following the consensus of three major algorithm groups: (i) presence only, (ii) presence and absence, and (iii) presence and background [27]. Based on these three categories, four algorithms were applied to generate the ENMs: (1) Bioclim [62], a presence-only algorithm; (2) random forest [63]; (3) support vector machine (SVM) [64], both presence and pseudoabsence algorithms; and (4) Maxent (Maxent 3.4.1) [65], which is considered a presence and background algorithm. To assess model performance, we used two evaluation metrics: the area under the curve (AUC) [66] and true skill statistics (TSS) [67]. All algorithms and performance metrics were implemented via the *ENMTML* R package [51].

For the postprocessing phase, ensemble models [68] were generated based on the suitability outputs. This approach supports more robust decision-making by combining the strengths of individual models, improving reliability for planning purposes [68]. To be included in the ensemble, models were required to meet the criteria of an AUC > 0.7 or a TSS > 0.5. The "MEAN" and "PCA" ensemble methods were applied using the *ENMTML* R package [51]. The final step involved calculating a threshold to convert continuous suitability values into binary predictions, using the method that maximizes the sum of sensitivity and specificity (thr = c(type = 'MAX_TSS')) [69]. These thresholds were used to generate binary maps indicating the potential presence (1) or absence (0) of each species. The intersection of binary maps was used to identify areas of potential hybridization between species [70]. All geoprocessing and mapping were conducted in QGIS Desktop 3.22.10 [71], and all analyses were performed using R 4.3.0 [72].

Results

Euterpe oleracea × *Euterpe edulis* – genomic diversity and population structure

In dataset 1, 393,263 loci were genotyped across 113 individuals of *E. edulis*, *E. oleracea*, and their hybrids. The mean sample coverage was 13.5× (SD = 3.7×, min = 5.9×, max = 27.1×), and the average number of sites per locus was 101.0. After filtering, 27,411 high-quality SNPs were retained. The mean proportion of missing data per individual was 0.0559 ± 0.093. Compared with both parental species, the hybrids presented the highest genomic diversity estimates (Table 1). For *E. edulis*, the observed heterozygosity (H_O) ranged from 0.078 to 0.149, and the gene diversity (H_S) ranged from 0.076 to 0.136. Individuals from the IAC Ubatuba (Ubatuba, SP) presented the second-highest diversity estimates and the highest number of private alleles (*Ap*). The lowest values of H_O (0.078), number of alleles (*A;*18.543) and A_P (55) were recorded at Ilha do Cardoso.

E. oleracea presented lower genomic diversity overall, with H_O values ranging from 0.050 to 0.051 and H_S values ranging from 0.043 to 0.046. Inhangapi (PA) had the highest number of *A* (30.985), whereas Peixe-Boi (PA) had the highest number of A_P (133). The lowest *A* and A_P values were found in Acará (30,801) and São Francisco do Pará (62), respectively. All groups, including hybrids, presented negative fixation (*f*).

According to Wright's F-statistics, most of the structuring was attributed to differentiation among populations, as indicated by a high F_{ST} value (F_{ST} = 0.469; F_{IS} = −0.098, F_{IT} = 0.446). The pairwise F_{ST} values revealed that the *E. oleracea* × *E. edulis* hybrids were genomically closer to the *E. oleracea* individuals, with F_{ST} values ranging from 0.286 to 0.296 between these two groups (Fig. 2). In contrast, the genetic distance between hybrids and *E. edulis* individuals ranged from 0.443 to 0.519. Among the *E. edulis* sites, the lowest F_{ST} values (0.052) were observed between Fazenda Capricórnio (Ubatuba, SP) and Santa Virgínia (São Luiz do Paraitinga, SP), while the highest values (0.120) were between Fazenda Capricórnio and Ilha do Cardoso. For *E. oleracea*, the lowest F_{ST} (0.042) was found between Inhangapi and São Francisco do Pará, and the highest (0.107) was found between Peixe-Boi and Acará. As expected, *E. edulis* and *E. oleracea* presented high pairwise F_{ST} values (> 0.820), indicating strong genetic differentiation between the species.

PCA (Fig. 3a) explained 55.68% of the total variation in the first two components (Supplementary Fig. 2). Most *E. oleracea* × *E. edulis* hybrids were positioned between the parental species in the scatterplot, although one individual (holeraceaedulis14) clustered closer to *E. edulis*, while five others (holeraceaedulis1, 6, 2, 3, and 8) grouped near *E. oleracea*. DAPC with *K-means* identified five genomic clusters (*K* = 5; Supplementary Figs. 3a–d). Groups 1, 3, and 4 included only *E. edulis* palms. Cluster 1, although composed of *E. edulis* palms, was distinct from the other two clusters and was formed by individuals from Ilha do Cardoso (SP), which may reflect the geographic isolation of this site (Fig. 3b). Group 2 was formed by *E. oleracea* and the five hybrids positioned near this species in the PCA. Group 5 comprised the remaining hybrids and one *E. edulis* individual from IAC Ubatuba (IACUbatuba5; Figs. 3b and c).

In the sNMF clustering (Fig. 3 d; Supplementary Fig. 4), four genomic (*K* = *4*) groups were detected, with stronger signals of introgression between *E. oleracea* and *E. edulis*. Group 1 was composed mainly of *E. edulis*, but

Table 1 Genomic diversity estimates for 43 *E. oleracea* individuals, 16 *E. oleracea* × *E. edulis* hybrids, and 54 *E. edulis* individuals, sampled across four sites in Pará and six sites in São Paulo, based on 27,411 SNP markers

Site	Code	Species	*N*	*HO*	*HS*	*A*	*AR*	*AP*	*f*
Cananeia (SP) - Ilha do Cardoso	ICardoso	*E. edulis*	4	0.078	0.081	18,543	1.182	55	−0.005 (0.016; 0.065)
São Luiz do Paraitinga (SP) - Santa Virgínia	SVirginia	*E. edulis*	16	0.099	0.097	37,501	1.221	2,111	−0.008 (−0.038; −0.021)
Ubatuba (SP) - Casa da Farinha	CFarinhaUbatuba	*E. edulis*, introduction of *E. oleracea*	10	0.077	0.076	33,926	1.174	351	0 (−0.027; −0.003)
Ubatuba (SP) - Fazenda Capricórnio	FCapricornioUbatuba	*E. edulis*, introduction of *E. oleracea*	11	0.081	0.077	34,465	1.173	201	−0.026 (−0.060; −0.037)
Ubatuba (SP) - IAC Ubatuba	IACUbatuba	*E. edulis*, introduction of *E. oleracea*	13	0.149	0.139	47,498	1.336	2,531	−0.022 (−0.084; −0.070)
Campinas (SP) - Instituto Agronômico	holeraceaedulis	Hybrid - *E. oleracea* × *E. edulis*	16	0.354	0.289	48,107	1.607	430	−0.169 (−0.229; −0.222)
Acará (PA)	Acara	*E. oleracea*	9	0.05	0.044	30,801	1.097	105	−0.094 (−0.153; −0.114)
Inhangapi (PA)	Inhangapi	*E. oleracea*	13	0.049	0.043	30,985	1.094	107	−0.077 (−0.152; −0.116)
Peixe-Boi (PA)	PBoi	*E. oleracea*	11	0.05	0.044	30,854	1.096	133	−0.078 (−0.154; −0.118)
São Francisco do Pará (PA)	SFranciscodoPara	*E. oleracea*	10	0.051	0.046	30,949	1.1	62	−0.071 (−0.132; −0.095)

N = Number of individuals, HO = Observed heterozygosity; HS = Gene diversity; A = Total number of alleles; AR = Allelic richness; AP = Number of private alleles; f = Fixation index (Confidence intervals)

introgression was detected in hybrids (holeraceaedulis9) and, at low proportions, in *E. oleracea* individuals across all sites (e.g. Acara2). A similar pattern was observed in Group 4. Group 2 included mostly *E. oleracea* palms but presented signs of *E. edulis* introgression, particularly in individuals from Ubatuba (SP), where historical introductions of *E. oleracea* have been documented. Interestingly, Group 2 also included individuals from Ilha do Cardoso (SP), where no such introductions have been reported. Group 3 included mostly hybrids and contained introgression from *E. edulis*. Importantly, when $K=2$, the hybrids that composed Group 3 at $K=4$ showed the expected admixed ancestry between the two parental species, whereas at higher K values they tended to form a distinct cluster, with subtle signals of introgression becoming apparent between species (Supplementary Fig. 5).

Euterpe precatoria × *Euterpe oleracea* – Genomic diversity and population structure

In dataset 2, a total of 672,257 loci were genotyped across 212 palms of *E. oleracea*, *E. precatoria*, and putative *E. precatoria* × *E. oleracea* hybrids. The mean coverage per sample was 12.8× (SD = 2.9×, min = 7.4×, max = 28.9×), and the average number of sites per locus was 101.0. The mean proportion of missing data per individual was 0.0384 ± 0.029. After the filters were applied in the *populations* module, 27,346 high-quality SNP markers were retained. As observed in dataset 1, the hybrids presented the highest genomic diversity estimates (Table 2). For *E. oleracea*, the observed heterozygosity (H_O) ranged from 0.057 to 0.068, which was consistently higher than that of H_S (0.050–0.056). The Itacoatiara (AM) population presented the highest A, A_R, and A_P, whereas the Peixe-Boi (PA) population presented the lowest A (30,742), the Inhangapi (PA) had the smallest A_R (1,117), and the Acará (PA) population presented no A_P. In *E. precatoria*, the H_O ranged from 0.046 to 0.111 and exceeded the H_S in all cases (0.036–0.107). The Presidente Figueiredo (AM) site presented the highest H_O, H_S, A, and A_R, whereas the lowest H_O and H_S were found in Itacoatiara and Parintins (AM; 0.046 and 0.036, respectively), Presidente Figueiredo had the lowest A_P (19). While sites in Pará had the lowest estimates of genetic diversity and number of private alleles for *E. oleracea*, populations in Amazonas generally presented greater diversity, particularly for *E. precatoria*.

Wright's F-statistics indicated that population subdivision contributed substantially to genetic structure (F_{ST} = 0.688; F_{IS} = −0.346, F_{IT} = 0.654), like dataset 1. Pairwise F_{ST} revealed that hybrids presented intermediate genomic distances from both parental species (*E.*

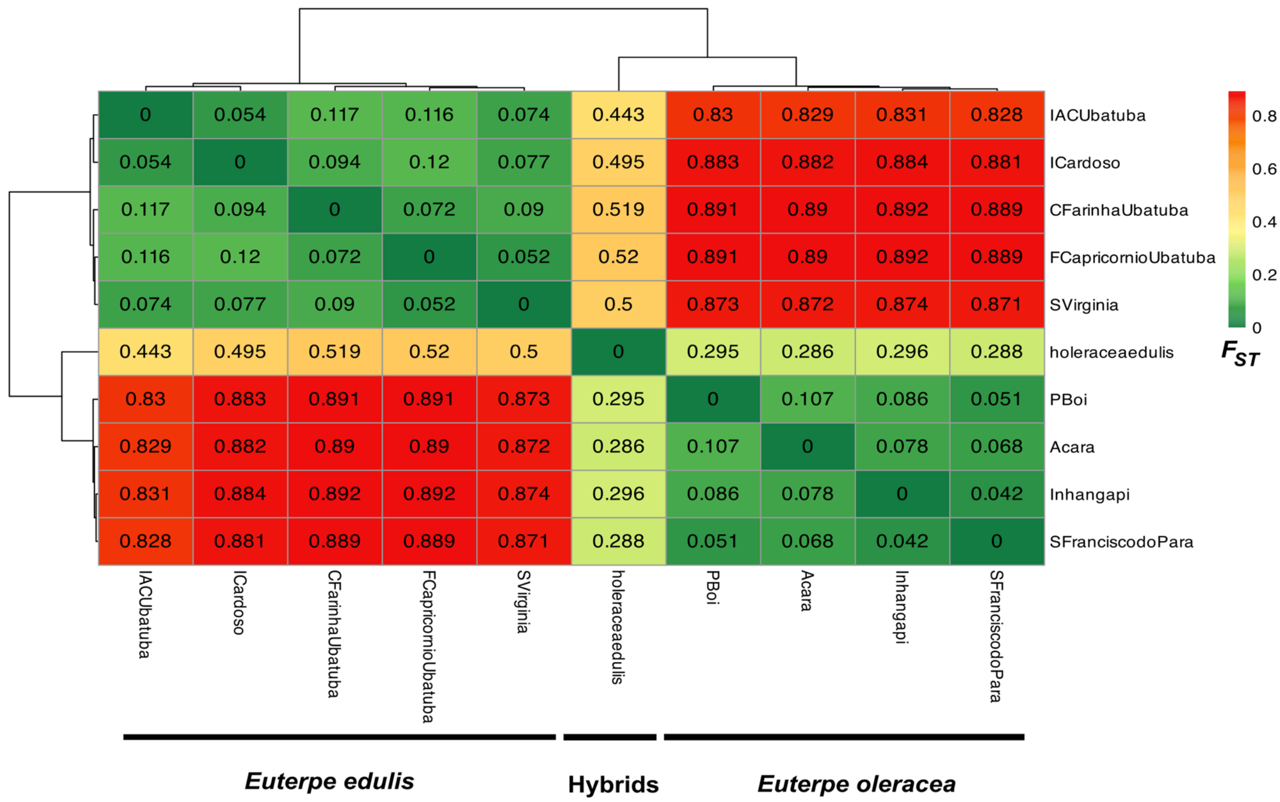

Fig. 2 Within-cell pairwise F_{ST} values (Nei 87) and F_{ST} heatmap based on collection sites considering 43 *E. oleracea* palms, 16 *E. oleracea* × *E. edulis* hybrids and 54 *E. edulis* palms, with 27,411 SNPs

oleracea: 0.454; *E. precatoria*: 0.484). The lowest divergence between hybrids and *E. oleracea* occurred in Parintins (AM; 0.448), and with *E. precatoria* occurred in Presidente Figueiredo (AM; 0.420). The degree of genomic differentiation between the *E. oleracea* and *E. precatoria* collection sites was high (F_{ST} >0.87; Fig. 4). Among the *E. oleracea* populations, the F_{ST} ranged from 0.050 (Peixe-Boi–São Francisco do Pará) to 0.221 (Inhangapi–Itacoatiara). For *E. precatoria*, values ranged from 0.007 (Presidente Figueiredo–Anori) to 0.402 (Manicoré–Parintins).

In the PCA (Fig. 5a), the first two PCs explained 67.59% of the total variation (Supplementary Fig. 6). The putative *E. precatoria* × *E. oleracea* hybrids were positioned between the parental species, with only one *E. precatoria* individual (PFigueiredo2_9) clustering nearby (Fig. 5a). DAPC via *K-means* identified eight genomic groups (*K* = *8*; Supplementary Figs. 7a-d). Groups 1, 5, 7, and 8 were exclusively composed of *E. precatoria* palms (Figs. 5b and c), while Groups 3, 4, and 6 consisted of *E. oleracea* individuals. Group 2 included the hybrids and the same *E. precatoria* individual (PFigueiredo2_9) found near them in the PCA.

In the sNMF analysis, seven genomic groups were detected (*K* = *7*; Fig. 5d; Supplementary Fig. 8). Groups 1, 5, and 7 were composed mainly of *E. oleracea*, while Groups 2, 4, and 6 consisted of *E. precatoria*. Group 3 included all the hybrids and PFigueiredo2_9. Moreover, sNMF was the only method used to clearly detect introgression between species. Small proportions of *E. oleracea* genomic material (Group 5) were detected in *E. precatoria* individuals from Anori and Itacoatiara (AM). Additionally, Group 7 included *E. precatoria* palms from Itacoatiara, and Group 1 appeared in individuals from Itacoatiara, Manicoré, Novo Airão, and Presidente Figueiredo (AM). The hybrid holeraceaprecatoria3 showed introgression from Group 6, which was associated with *E. precatoria*. As in the previous dataset, hybrids displayed admixed ancestry at *K* = 2 with approximately equal (~ 50/50) ancestry from both parental species, while at higher *K* values they tended to form a distinct cluster (Supplementary Fig. 9).

Niche overlap

After herbarium database queries, quality filtering, and spatial rarefaction within modeling cells, a total of 327 occurrence points for *E. oleracea*, 283 for *E. edulis*, and 1,276 for *E. precatoria* were retained for niche overlap analysis and ecological niche modeling (ENM). The environmental variation was summarized through a PCA, with PC1 explaining 44.69% and PC2, 18.47% of the total variance. These principal components were used to

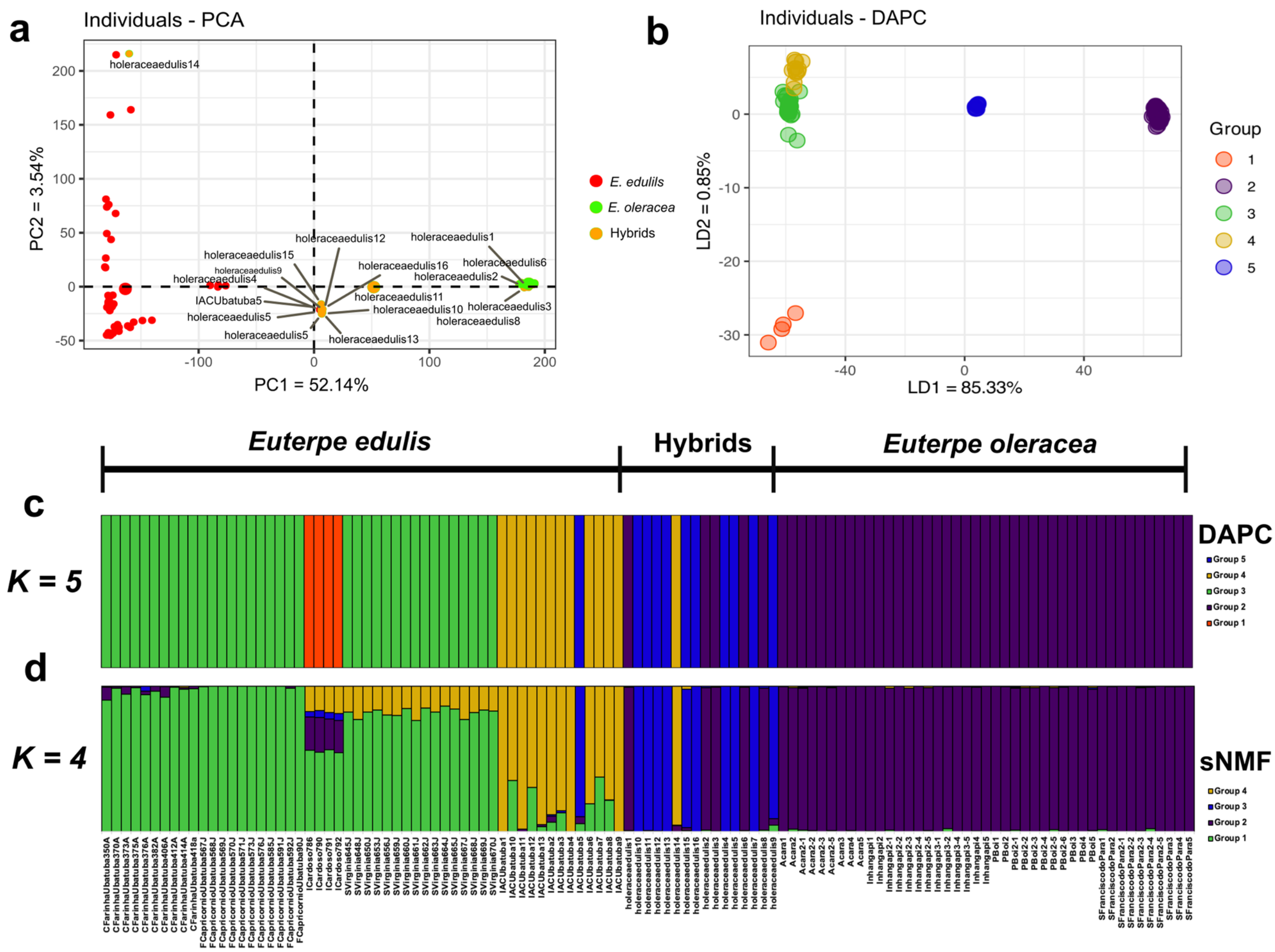

Fig. 3 Genetic structure groups based on collection sites considering 43 *E. oleracea* palms, 16 *E. oleracea* × *E. edulis* hybrids, and 54 *E. edulis* palms, with 27,411 SNPs. **a** Distribution of genomic groups defined by PCA; **b** distribution of genomic groups defined by DAPC *K-means*, with *K* = 5; **c** bar plots with the DAPC by *K-means* ancestry coefficients, considering *K* = *5*; **d** sNMF ancestry coefficients, for *K* = *4*

quantify the niche overlap between congeneric species (Fig. 6a and b). According to Schoener's D index, *E. oleracea* and *E. edulis* presented low niche overlap (D = 0.139), whereas *E. oleracea* and *E. precatoria* showed substantial overlap (D = 0.726). Despite the low overlap between *E. oleracea* and *E. edulis*, there was a high potential for niche expansion (71.68%, red region) and a considerable proportion of unfilled shared niches (49.70%, green region; Fig. 6a). In contrast, *E. oleracea* and *E. precatoria* presented high niche stability in their area of overlap (94.45%, blue region; Fig. 6b).

Ecological niche modeling and potential hybridization zones

The final ecological niche models exhibited high performance, with AUC values ranging from 0.928 to 0.996 and TSS between 0.705 and 0.952 (Supplementary Table 1). Given the similar results obtained by both ensemble approaches (mean and PCA-based), we used the averaging method.

According to the ecological niche models, the Atlantic Forest biome showed the highest environmental suitability for *E. edulis*, although suitable areas were also identified in other Brazilian biomes, such as the Cerrado and the Amazon. Some Latin American countries, including Venezuela, Guyana, and Colombia, also showed high suitability, despite the absence of occurrence records (Fig. 6c). For *E. oleracea*, the Amazon biome presented the broadest suitability, with a potential distribution from east to west, as well as in coastal areas of the Atlantic Forest (Fig. 6d). This species also showed high suitability in neighboring South American countries and parts of Central America, such as Panama and Costa Rica. *E. precatoria* displayed a suitability pattern similar to that of *E. oleracea*, although with broader areas of high suitability, especially in the western Amazon (Fig. 6e).

Using the ensemble method (MEA), binary maps were generated to define suitable areas for each species (1 = suitable, 0 = unsuitable). From these maps, we identified potential hybridization zones, i.e., areas

Table 2 Genomic diversity identified in 93 *E. oleracea* individuals, five *E. precatoria* × *E. oleracea* hybrid accessions, and 114 *E. precatoria* individuals, sampled from 13 collection sites in the state of Amazonas and four in Pará, based on 27,346 SNP markers

Localities	Code	Species	*N*	*HO*	*HS*	*A*	*AR*	*AP*	*f*
Acará (PA)	Acara	*E. oleracea*	9	0.06	0.052	30,866	1.122	0	−0.112 (−0.178; −0.138)
Inhangapi (PA)	Inhangapi	*E. oleracea*	13	0.057	0.05	31,167	1.117	20	−0.095 (−0.177; −0.137)
Itacoatiara (AM)	Itacoatiara	*E. oleracea*	18	0.066	0.055	32,011	1.129	957	−0.133 (−0.226; −0.195)
Nova Airão (AM)	NAirao	*E. oleracea*	10	0.068	0.053	31,018	1.122	227	−0.203 (−0.296; −0.262)
Parintins (AM)	Parintins	*E. oleracea*	10	0.059	0.056	31,075	1.135	52	−0.015 (−0.071; −0.031)
Peixe-Boi (PA)	PBoi	*E. oleracea*	11	0.058	0.05	30,742	1.12	9	−0.088 (−0.170; −0.130)
Presidente Figueiredo (AM)	PFigueiredo	*E. oleracea*	12	0.062	0.051	31,540	1.122	35	−0.142 (−0.223; −0.193)
São Francisco do Pará (PA)	SFranciscodoPara	*E. oleracea*	10	0.06	0.053	31,371	1.126	21	−0.087 (−0.160; −0.120)
Manacapuru (AM)	Hprecatoriaoleracea	*Hybrids - E. precatoria* × *E. oleracea*	5	0.641	0.352	45,165	1.735	1,108	−0.782 (−0.828; −0.818)
Anori (AM)	Anori	*E. precatoria*	13	0.076	0.067	32,936	1.16	34	−0.080 (−0.151; −0.121)
Borba (AM)	Borba	*E. precatoria*	14	0.063	0.056	31,670	1.131	532	−0.071 (−0.140; −0.105)
Codajás (AM)	Codajas	*E. precatoria*	15	0.073	0.069	33,181	1.165	126	−0.017 (−0.082; −0.054)
Itacoatiara (AM)	Itacoatiara2	*E. precatoria*	13	0.046	0.036	29,951	1.089	272	−0.166 (−0.300; −0.253)
Manicoré (AM)	Manicore	*E. precatoria*	13	0.071	0.06	32,067	1.142	1,496	−0.118 (−0.194; −0.163)
Nova Airão (AM)	NAirao2	*E. precatoria*	18	0.07	0.059	32,158	1.141	757	−0.107 (−0.186; −0.157)
Parintins (AM)	Parintins2	*E. precatoria*	14	0.047	0.036	30,294	1.083	1,211	−0.206 (−0.344; −0.298)
Presidente Figueiredo (AM)	PFigueiredo2	*E. precatoria*	14	0.111	0.107	46,219	1.316	19	0.012 (−0.049; −0.029)

N = Number of individuals, HO = Observed heterozygosity; HS = Gene diversity; A = Total number of alleles; AR = Allelic richness; AP = Number of private alleles; f = Fixation index (Confidence intervals)

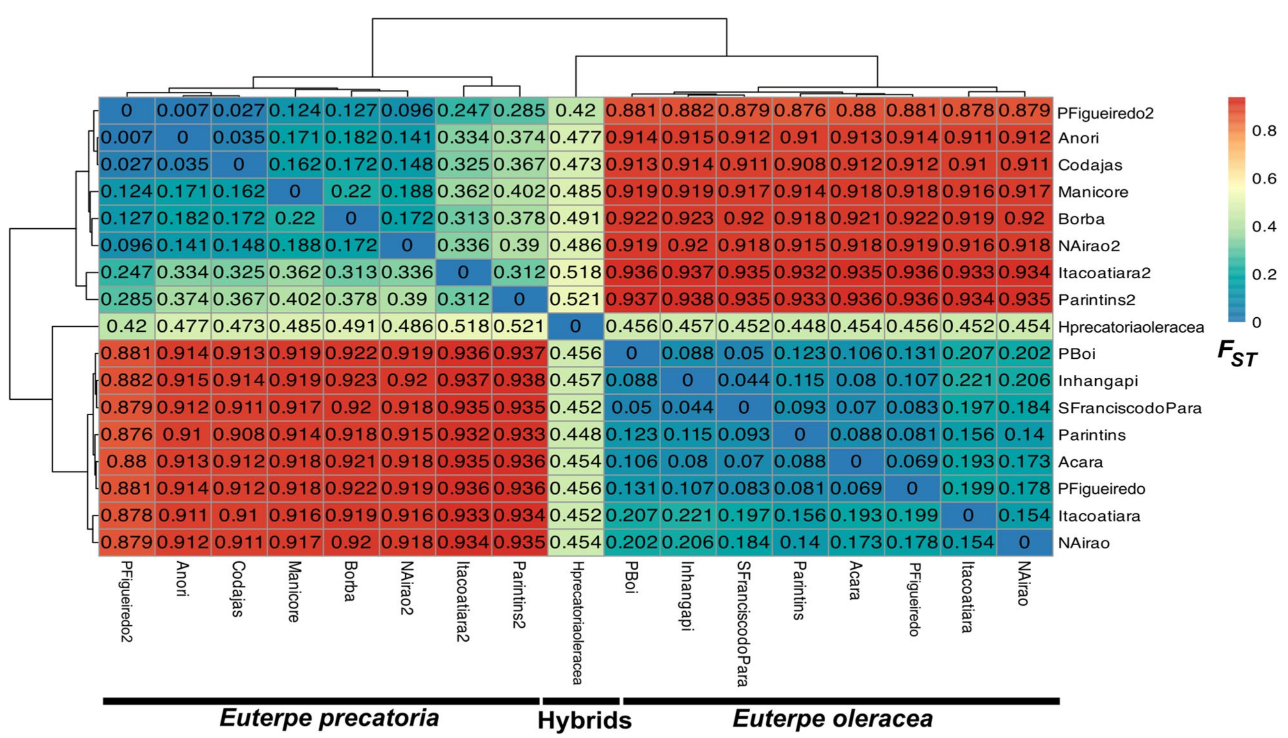

Fig. 4 Within-cell pairwise F_{ST} values and F_{ST} heatmap based on collection sites considering 93 *E. oleracea* palms, five accessions of *E. precatoria* × *E. oleracea* hybrids and 114 *E. precatoria* palms, with 27,346 SNP markers

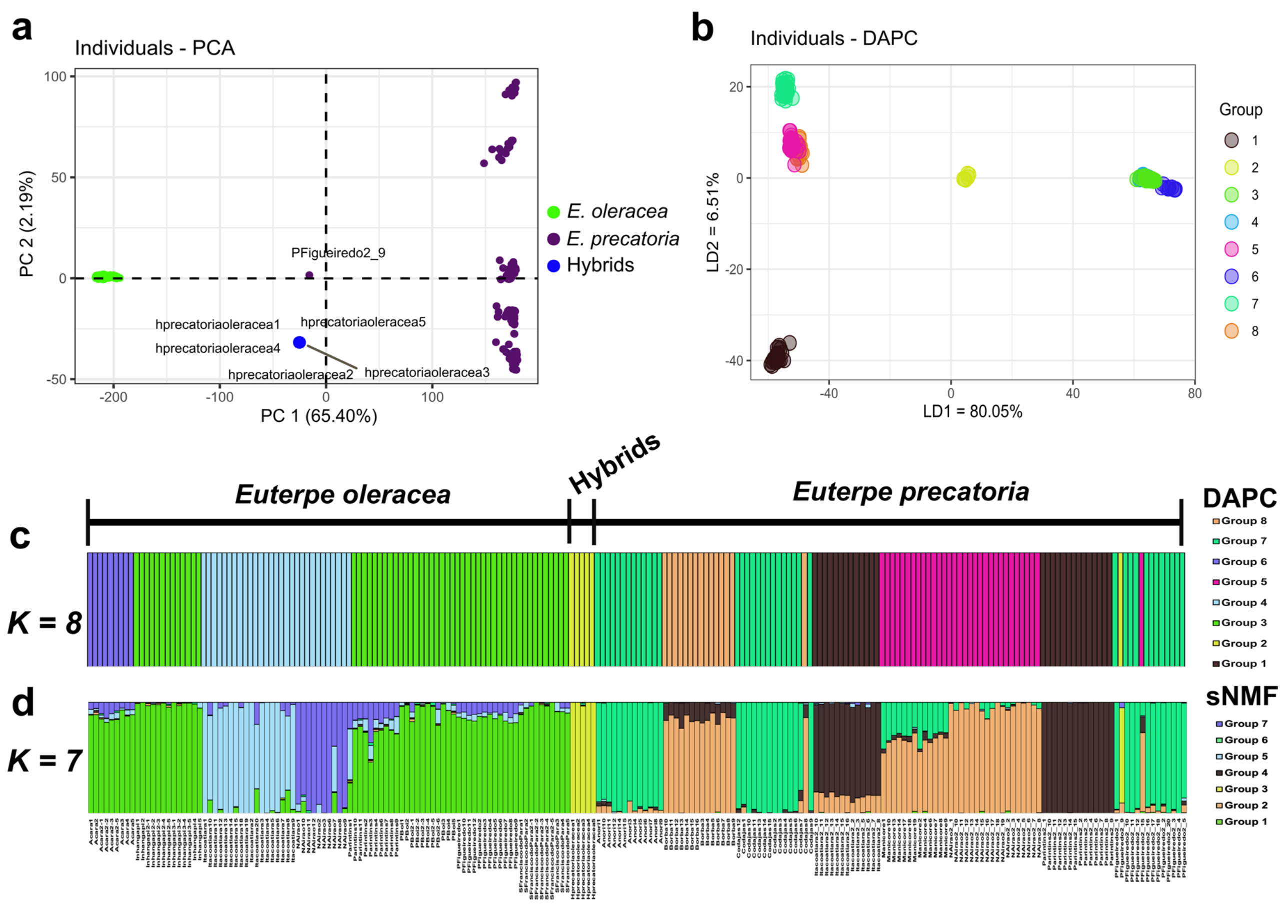

Fig. 5 Genetic structure groups based on collection sites considering 93 *E. oleracea* palms, five accessions of *E. precatoria* × *E. oleracea* hybrids, and 114 *E. precatoria* palms, with 27,346 SNP markers. **A** Distribution of genomic groups defined by PCA; **B** distribution of genomic groups defined by DAPC *K-means*, with *K* = 8; **C** bar plots with the DAPC by *K-means* ancestry coefficients, considering *K* = 8; **D** sNMF ancestry coefficients, for *K* = 7

environmentally suitable for the co-occurrence of both species. Between *E. oleracea* and *E. edulis*, 2,223,467 km² were identified as potential hybridization zones. In the Atlantic Forest biome—where *E. oleracea* has been introduced for ornamental and cultivation purposes—81,466 km² were suitable for both species (Fig. 6f). For *E. oleracea* and *E. precatoria*, the extent of potential hybridization zones was greater, totaling 7,187,019 km². Within the Amazon, 5,956,366 km² were identified as environmentally suitable for both species, suggesting that most of the potential for hybridization between them is concentrated in this biome (Fig. 6g).

Discussion

DNA-based analyses have significantly advanced our understanding of hybridization processes in plants [73]. In palms of high economic value, such as oil palm (*Elaeis guineensis*) and date palm (*Phoenix dactylifera*), both mediated [74] and natural [73] hybridization and introgression have been reported. In *P. dactylifera*, these processes played a central role in shaping genome-wide patterns of diversity [73], while in *E. guineensis* × *E. oleifera* hybrids, high variability was observed in traits such as heritability and coefficients of variation [74]. In this study, we present evidence of genomic diversity and population structure supporting the occurrence of hybrids and introgression between *Euterpe oleracea* and other *Euterpe* species. We also explored the dynamics of ecological niches and potential co-occurrence zones to better understand how these processes may persist in natural and human-influenced landscapes.

Interspecific hybridization and introgression in *Euterpe* species

Euterpe oleracea × *E. edulis*

Despite the allopatric speciation of *E. edulis* [16], natural gene flow between these congeneric species remains possible. Observations by Tiberio et al. [4] revealed that their fruiting seasons overlap: *E. edulis* fruits from March to September and *E. oleracea* fruits from June to September. This overlap leads to interactions between *E. oleracea* and bird communities that disperse *E. edulis*. The presence of natural hybrids between the two species suggests

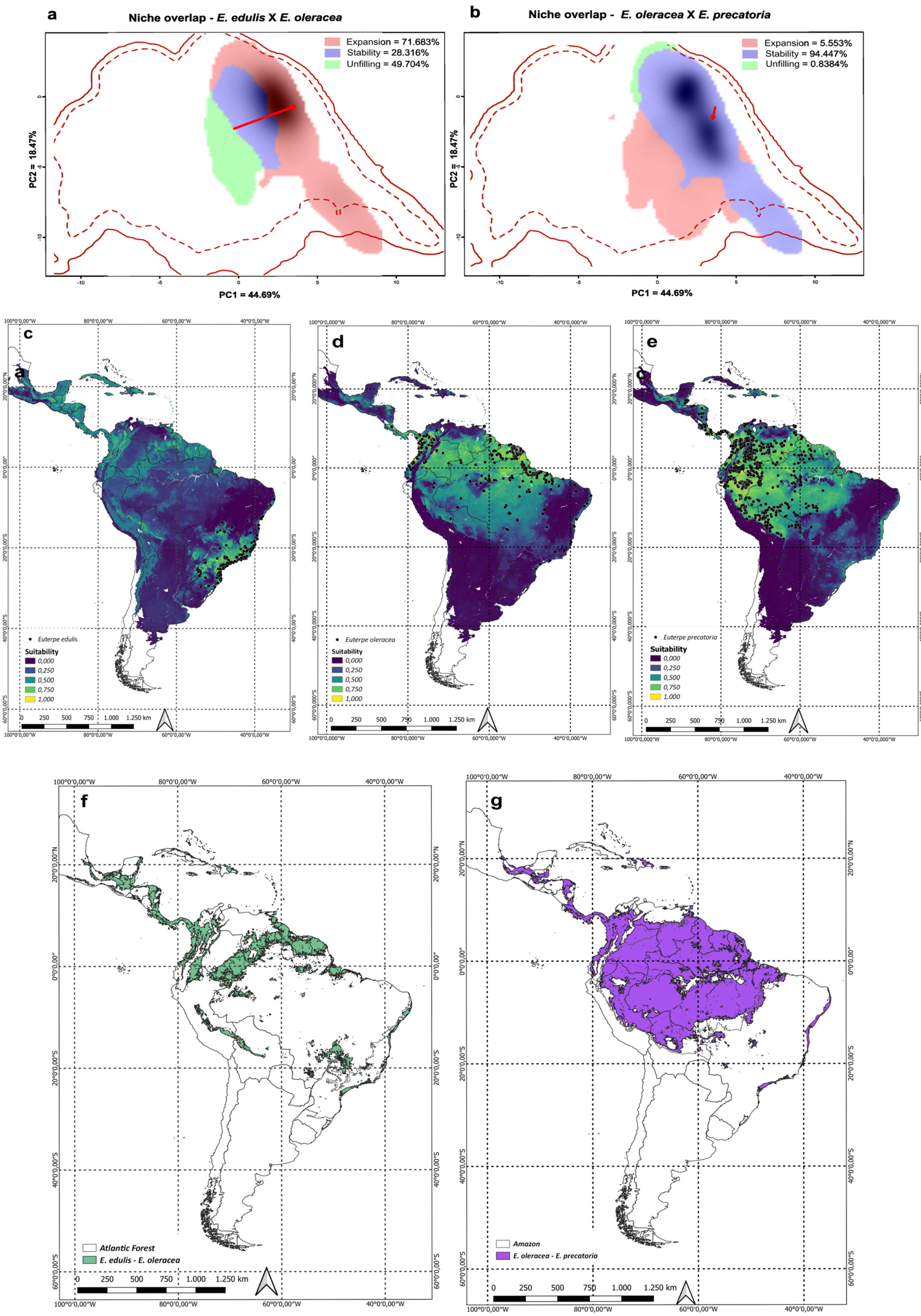

Fig. 6 (See legend on next page.)

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Fig. 6 Ecological niche overlap and modeling and overlapping suitable areas of *Euterpe* species. Niche overlap via Schoener's D metric: (**a**) between the species *E. edulis* × *E. oleracea* and (**b**) between *E. oleracea* × *E. precatoria*. Ensemble with the means of the algorithms used to model the species of the genus *Euterpe*: (**c**) map of the suitability distribution of *E. edulis* across Latin America, (**d**) map of the suitability distribution of *E. oleracea*, and (**e**) map of the suitability distribution of *E. precatoria*. Overlap of binary maps with suitable areas of *Euterpe* species: (**f**) between *Euterpe edulis* and *E. oleracea* and (**g**) between *E. oleracea* and *E. precatoria*. The solid and dashed lines in the letters a and b correspond to 100% and 50%, respectively, of the viability of each environment for the species, and the colors in these figures indicate the following: red = niche expansion, blue = niche stability, and green = unfilled niche. The red arrow in the letter represents the change in the centroid of the niche from one species to another

that pollen exchange may occur as a result of simultaneous flowering [4].

We detected a high level of genomic diversity in the *E. oleracea* × *E. edulis* hybrids derived from cross-mediated breeding [3]. High genetic diversity is recognized as a strong indicator of hybridization in plants [24, 75]. Most of these hybrids also presented intermediate positions between the parental species in the population structure analyses, providing clear evidence of successful interspecific crossing. Additionally, clustering methods revealed the formation of a nearly exclusive genomic group representing the hybrids. Considering $K = 2$ in the sNMF analysis, hybrids showed the expected admixed ancestry. Similar outcomes have been reported in simulation studies, where populations originating from past admixture are detected as admixed when $K = 2$, but prolonged drift after the hybridization event may lead them to appear as an independent cluster at higher K values [76]. However, five individuals identified as hybrids shared the same genomic group as *E. oleracea* did, suggesting a closer genetic affinity to this species. A similar pattern has been observed in *Quercus* spp., where parental species do not contribute equally to hybrid genotypes [77]. Nonetheless, it remains possible that these individuals classified as hybrids are, in fact, pure *E. oleracea* palms. In addition, although we did not have access to natural hybrid material [4], their inclusion in future studies would provide an important opportunity to expand these findings.

Beyond hybridization, signs of introgression were also detected between *E. oleracea* and *E. edulis*, indicating the incorporation of genetic material from one taxon into the gene pool of the other in an asymmetric admixture process [1]. Introgressions were observed at collection sites in the municipality of Ubatuba and on Ilha do Cardoso, in Cananeia (SP). Notably, there are no documented records of *E. oleracea* introductions on Ilha do Cardoso. Although only four palms were sampled in this location, this raises the hypothesis that their genetic composition may result from introductions in nearby regions, such as Sete Barras and Pariquera-Açu (SP), approximately 120 km and 60 km away, respectively. According to Gaiotto et al. [11], paternity analyses have revealed unexpected long-distance dispersal (up to 22 km) in *E. edulis*, which is likely mediated by wind and birds such as toucans (Ramphastidae). The predominance of wind in *E. edulis* seed dispersal was recently confirmed by Zamudio et al. [78]. In contrast, no evidence has linked *E. oleracea* seed dispersal to wind. In its native environment, *E. oleracea* is pollinated by a wide array of insects, including bees, flies, beetles, wasps, and ants [79], and its seeds are typically dispersed over short distances by small mammals and over longer distances by birds, rivers, and human activities [80]. However, since *E. oleracea* has already been observed to interact with bird species that disperse *E. edulis* in the Atlantic Forest [4], it is plausible that new ecological behaviors may be emerging for *E. oleracea* in this biome.

Euterpe precatoria × *E. oleracea*

In swampy and disturbed areas, *E. oleracea* can behave as an aggressive colonizer [13]. Owing to the growing demand for açai fruits, the cultivation of *E. oleracea* has expanded in the state of Amazonas [15, 21], a region in the western Amazon where the species did not originally occur [13]. According to Pichardo-Marcano et al. [16], *E. oleracea* diverged earliest from the other five *Euterpe* species, approximately 8.04 Mya during the Miocene. However, a phylogeny based on chloroplast genomes suggests that *E. oleracea* and *E. precatoria* are sister species [81]. Despite the incongruences between nuclear and plastid phylogenies in *Euterpe*, our results show that natural hybrids between *E. precatoria* and *E. oleracea* can occur when the species are found in close proximity.

Therefore, as observed in *E. oleracea* × *E. edulis* hybrids, the highest genomic diversity was found in the accessions of putative *E. precatoria* × *E. oleracea* hybrids. These individuals also occupied an intermediate position between the parental species in the population structure analyses, indicating similar genomic distances to both *E. oleracea* and *E. precatoria*. Similarly, clustering methods revealed a nearly exclusive group corresponding to the hybrids. In the sNMF analyses, hybrids initially exhibited a balanced ancestry from both parental species at $K = 2$. As the number of clusters increased, however, they progressively separated into a distinct genomic group (Supplementary Fig. 9), as observed in *E. oleracea* × *E. edulis* [76]. Finally, at the sampling sites where both species co-occur, gene flow was detected through introgression events.

Overlaps in the flowering and fruiting periods between *E. oleracea* and *E. precatoria* have been observed in the western Amazon, although each species reaches its peak at different times. In Manaus, *E. oleracea* flowers predominantly from January to July (the rainy season), while *E. precatoria* flowers mostly from July to December (the dry season) [6]. However, both species presented a high presence of inflorescences from May to August. Fruit maturity peaks between October and December for *E. oleracea* and from May to July for *E. precatoria*, with both species showing a shared peak between May and June [6]. *E. oleracea* is pollinated by a wide variety of insects, including both specialists (e.g., curculionid beetles) and generalists [79, 82]. Studies by Campbell et al. [82] and Küchmeister et al. [83] indicate that beetle, fly, and bee families are common pollinators for both species. Seed and fruit dispersers are also common. For *E. oleracea*, long-distance dispersal is facilitated by birds such as toucans, guans, aracaris, parakeets, parrots, and thrushes, along with human and river transport [80]. *E. precatoria*, on the other hand, relies primarily on precipitation for seed dispersal [9] but also uses rivers and shares avian dispersers such as parrots and toucans [84]. The geographical proximity between the two species, enhanced by anthropic action, combined with overlapping reproductive phenology and shared pollinators and dispersers, likely contributed to gene flow. This is evidenced by the presence of both natural hybrids and signs of partial introgression.

Niche overlap and ecological niche modeling

The case of E. oleracea and E. edulis

Euterpe oleracea shows strong potential to expand into the Atlantic Forest biome, with 71.68% of *E. edulis*'s climatic niche representing suitable but uncolonized areas, and nearly 50% of their shared niche is still unfilled. This suggests a high likelihood of *E. oleracea* establishing in regions currently occupied by *E. edulis*, particularly along the entire coastal zone of the Atlantic Forest, as indicated by ecological niche modeling. Carvalho et al. [85] developed suitability models for *E. edulis* considering the Last Glacial Maximum (21 ka), the Middle Holocene (6 ka), and the present day (0 ka). Coastal areas have remained the most climatically stable over time, serving as important refugia for *E. edulis* [85]. In 2019, de Souza and Prevedello [86] reported that approximately 15% of the biome still contains areas suitable for *E. edulis*, especially in coastal regions and in the interior of Paraná state (dense and mixed ombrophilous forests). In 2020, a 39% niche overlap was identified between the Australian palm *Archontophoenix cunninghamiana*, which is invasive in the Atlantic Forest biome, and *E. edulis*. Although the invasion potential of this palm may be reduced under climate change scenarios, interactions with the frugivorous dispersers of *E. edulis* have already been observed [87]. Similar interactions have been identified between *E. edulis* and *E. oleracea* [4]. These processes may favor invasions, as similar ecological functions can be fulfilled by invasive species in place of native species [87]. Finally, in 2021, de Souza and Prevedello [88] modeled the climatic suitability of *E. oleracea* in areas currently occupied by *E. edulis* under both present and future climate scenarios. In this case, a more concerning invasion dynamic was identified: unlike the Australian palm, *E. oleracea*, especially under climate change, may further restrict the distribution range of *E. edulis* [88].

Our findings align with those of de Souza and Prevedello [88] given the great potential for niche expansion of *E. oleracea* toward *E. edulis*, as revealed by overlap analyses. Moreover, *E. oleracea* has a high capacity for resprouting, which makes its removal from the biome particularly challenging [4]. Genomic analyses confirmed a weak reproductive barrier between the species, allowing hybrid formation, with the potential for backcrossing with *E. edulis* and contributing to genetic erosion in the native species. Additionally, introgression of *E. edulis* genetic material into *E. oleracea* individuals was detected, which may further increase the adaptive capacity of *E. oleracea* through the incorporation of locally adapted genes from *E. edulis* [88]. Taken together, these factors support the classification of *E. oleracea* plantings and other occurrences in the Atlantic Forest as invasive, nonnative congeneric species [4, 88].

Based on the results and evidence presented here, several conservation practices can be proposed to support *E. edulis* and mitigate the impacts of *E. oleracea* expansion. First, Atlantic Forest remnants in coastal regions should be prioritized for *E. edulis* conservation, as these areas are the most climatically stable for the species and present the highest risk of uncontrolled *E. oleracea* invasion and hybrid proliferation. Second, it is necessary to promote the expansion of the *E. edulis* distribution especially along the coast and in the southern states of Brazil, where populations have been depleted but high suitability has been consistently indicated by the models. Finally, restoration programs in the Atlantic Forest and the promotion of frozen pulp production, very similar to that of Amazonian açaí (*E. oleracea*) [89], should be implemented to encourage the cultivation and use of *E. edulis*. Sustainable fruit harvesting involving traditional communities has proven effective in maintaining the genetic diversity of *E. edulis* [90, 91]. The species also benefits from its proximity to consumer markets and the extensive areas suitable for sustainable management [92].

The case of E. oleracea and E. precatoria

Unlike the interaction between *E. oleracea* and *E. edulis*, the açaí palms (*E. oleracea* and *E. precatoria*) exhibited a high degree of niche overlap. As both are native to northern Brazil [13], they are well suited to nearly the entire Amazon biome, which acts as a secondary contact zone. The ecological niche models developed by Moscoso et al. [93] and Vaz & Nabout [94] also indicated high environmental suitability for *E. oleracea* in the Amazon biome and along the northeastern coast of Brazil.

Neither of the two açai species (*E. oleracea* and *E. precatoria*) is currently considered threatened with extinction, unlike *E. edulis*. According to ter Steege et al. [95], in the Amazon biome, *E. precatoria* was identified as the most hyperdominant species, followed by *E. oleracea*, which is the fifth most hyperdominant. This means that both species exhibit wide geographic distributions and the highest relative abundances in the sampled plots [95]. The combination of high demand for açaí pulp, the hyperdominance of these palms, and their high environmental suitability in the Amazon makes genetic contamination of native *E. precatoria* populations by introduced *E. oleracea* plantations virtually inevitable under current conditions [15]. This scenario may increase gene introgression between congeneric species in areas of co-occurrence, leading to the formation of natural hybrids, as previously observed.

Since each of the açai species offers distinct productive advantages, generating hybrids between them is of agronomic interest [96]. Although limited information is available about cultivars developed from açaí hybrids, beyond the findings of this study and those reported by Tavares [21], the Brazilian Agricultural Research Corporation (Embrapa) successfully produced two *E. oleracea* × *E. precatoria* hybrids through controlled pollination in Belém, Pará [97]. These hybrids exhibit peak flowering from January to May, with fruiting occurring between May and November, and pollen viability ranging from ~ 79–81% in one hybrid to ~ 85–90% in the other, comparable to their parental species, indicating that they are fertile [97]. This reproductive behavior closely resembles that of *E. oleracea* cultivated in Manaus, Amazonas [15], as expected given that *E. oleracea* is the maternal species [97]. Nonetheless, even with the successful development of hybrid açai cultivars with superior traits, there remains significant concern regarding the potential for gene flow and subsequent genetic contamination of both native and natural populations, even though the number of individuals identified so far remains limited.

To preserve the genetic integrity of açai species, several management actions can be implemented. First, the wide genomic diversity identified within açai species [98] enhances the potential for developing new cultivars. Priority should be given to selecting individuals of *E. oleracea*, *E. precatoria*, and their hybrids (including reciprocal hybrids) that exhibit nonoverlapping flowering and fruiting periods to minimize gene flow. Second, it is essential to monitor and manage spontaneous occurrences of one species near plantations of another, as natural hybridization has been observed when *E. oleracea* accessions are located close to *E. precatoria* cultivation sites [21]. Finally, the cultivation of açai species should be aligned with their natural geographic distributions, for example, promoting the planting of *E. oleracea* in the eastern Amazon and *E. precatoria* in the western Amazon [13].

Conclusions

Despite the speciation observed within the *Euterpe* genus, our estimates of genomic diversity and population structure confirmed the occurrence of hybrids between *E. oleracea* × *E. edulis* and *E. precatoria* × *E. oleracea*. These hybrids presented high levels of genomic diversity and were structurally positioned as intermediaries between their parental species. Except for a few individuals among the *E. oleracea* × *E. edulis* hybrids—likely in need of reclassification—most formed distinct genomic groups. In addition to hybridization, we also identified introgression events between congeneric species. These incompatibility barriers are likely overcome by anthropogenic factors, particularly the expansion of *E. oleracea* plantations in the Amazon and Atlantic Forests. This co-occurrence of species increases the chances of gene flow, as they share flowering and fruiting periods, as do pollinators and dispersers.

These processes of hybridization and introgression are particularly concerning given the strong potential for *E. oleracea* to expand into the niche of *E. edulis*. Our ecological niche modeling highlighted the coastal Atlantic Forest biome as a priority area for the conservation of *E. edulis*. Promoting the management of this species by traditional communities may help mitigate some of the associated risks. Furthermore, the growing presence of *E. oleracea* plantations in the western Amazon could destabilize the ecological niches of both açaí species, raising concerns about the genetic contamination of natural populations. It is essential to expand research on cultivars whose flowering and fruiting times do not overlap to reduce gene flow and to prioritize the cultivation of each açai species within its native distribution range.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12870-025-07775-1.

Supplementary Material 1

Authors' contributions

AFF performed the population genomics and niche analyses and wrote the manuscript. MS filtered the raw data and obtained the SNP markers. JAMM conducted the sampling. JAMM and IASC prepared the GBS library. GOF extracted DNA from the samples. MHV obtained and curated the geographic coordinates for niche analyses. MSM contributed to structure analyses. JVRA processed the genotypic data. SLFR and VAM conducted the sampling. MTGL contributed to study design and provided financial support. MIZ conceived and coordinated the study and also provided financial support. All authors read, edited, and approved the final version of the manuscript.

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Data availability
The datasets generated and analyzed during the current study are available in the NCBI GenBank repository, https://www.ncbi.nlm.nih.gov/bioproject/PRJNA1289687. Scripts for niche overlap and ecological niche modeling, along with geographic coordinates, are available at GitHub: https://github.com/anaf-f/Euterpe_hybrids.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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